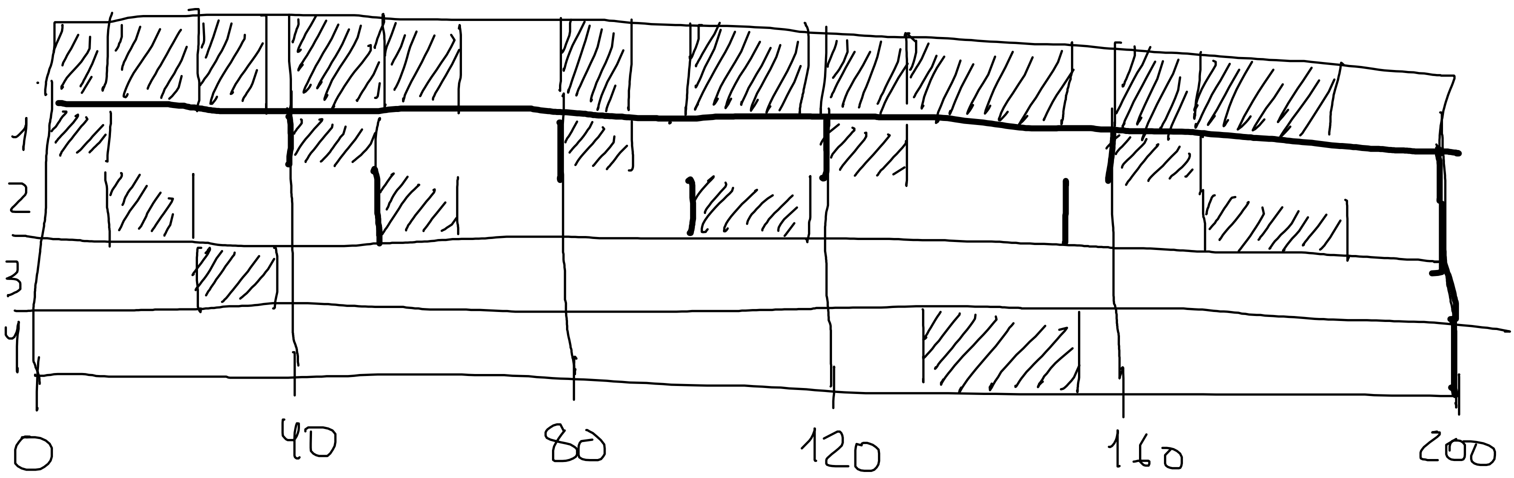
51. $T_m = mcm(40, 50, 200, 200) \to 200$

$\max(10, 18, 10, 20) \leq T_s \leq \min(40, 50, 200, 200)$

$20 \leq T_s \; 40 \qquad 200 \bmod T_s = 0$

$T_s \begin{cases} 20 \\ 25 \\ 40 \end{cases} \Big\} \; T_s \to 40$

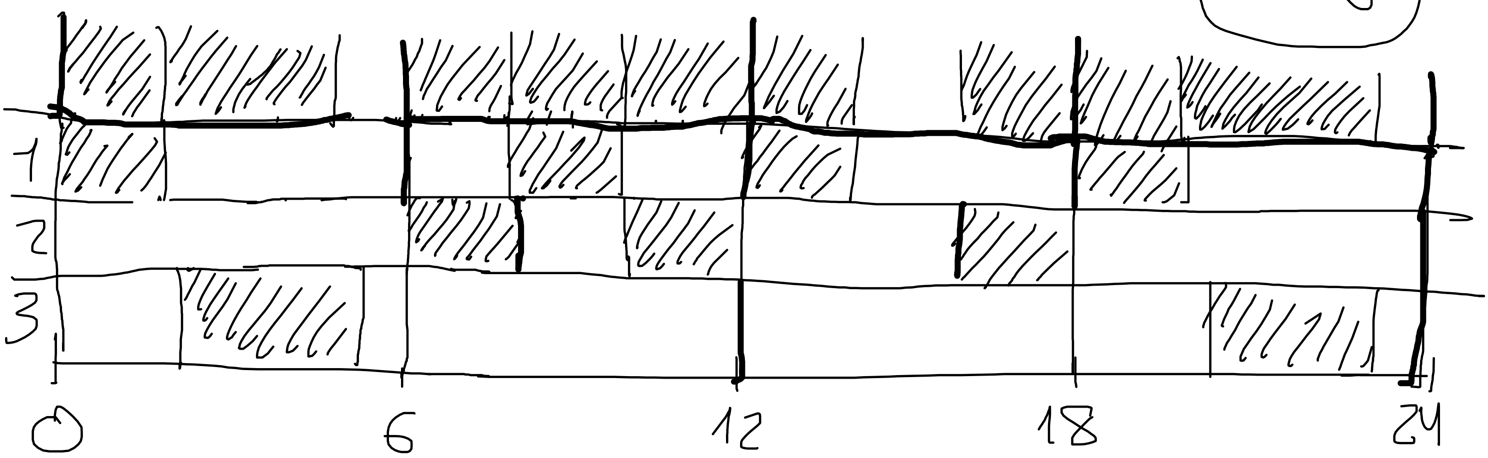


52.

$T_M = mcm(6, 8, 12) = 24$

$\max(2,2,3) \le T_S \le \min(6,8,12) \rightarrow 3 \le T_S \le 6$

$T_M \bmod T_S = 0 \rightarrow 24 \bmod T_S = 0 \rightarrow T_S \rightarrow 3, 4, 6 \rightarrow \boxed{T_S = 6}$



53.

$$U=\sum_{i=1}^{n}\frac{C_i}{T_i}=\frac{9}{30}+\frac{10}{40}+\frac{10}{50}=\frac{3}{10}+\frac{1}{4}+\frac{1}{5}=$$

$$=\frac{6+5+4}{20}=\frac{15}{20}=\frac{3}{4}=0'75$$

RMS $\rightarrow n(\sqrt[n]{2}-1) \rightarrow 3(\sqrt[3]{2}-1)=0'78$

$U<0'78 \rightarrow$ PLANIFICABLE

54.

$$U=\sum_{i=1}^{n}\frac{C_i}{T_i}=\frac{10}{40}+\frac{20}{60}+\frac{20}{80}=\frac{1}{4}+\frac{1}{3}+\frac{1}{4}=\frac{3+4+3}{12}=$$

$$=\frac{10}{12}=\frac{5}{6}=0'83$$

$U>0'78 \rightarrow$ no se puede afirmar ni negar que sea planificable

$\rightarrow$ no exacto ($U<1$)

$$T_M=mcm(40,60,80)=240$$

$$\underbrace{2^2\cdot 3\cdot 5 \quad 2^4\cdot 5}_{2^4\cdot 3\cdot 5}$$

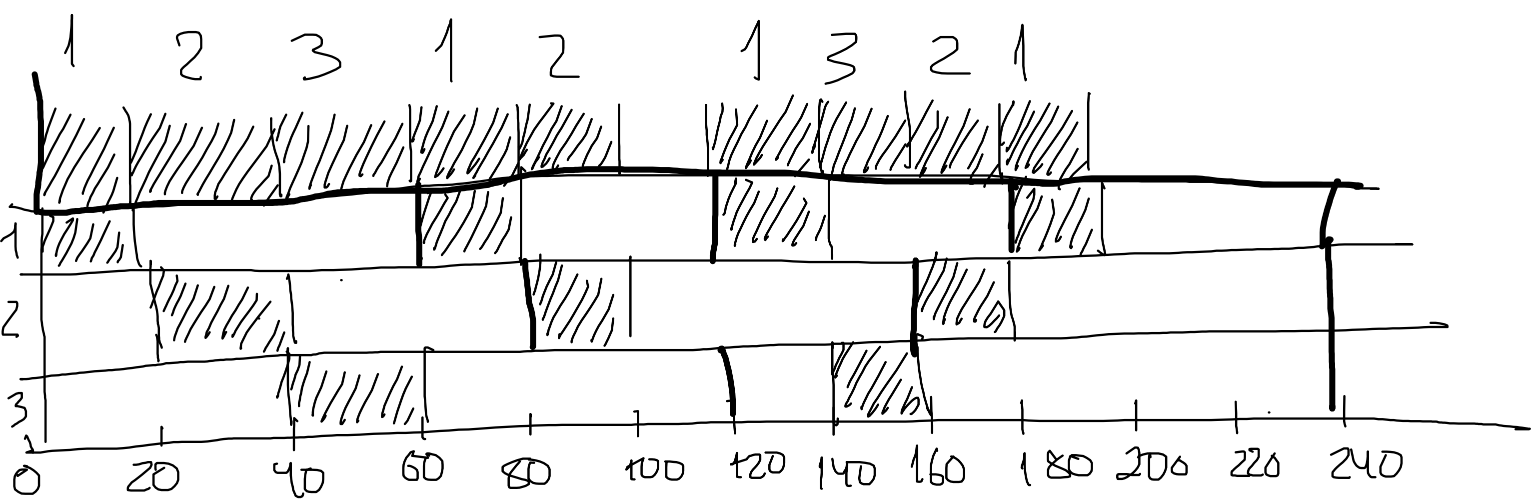
55

$$U = \frac{20}{60} + \frac{20}{80} + \frac{20}{120} = \frac{1}{3} + \frac{1}{4} + \frac{1}{6} = \frac{4+3+2}{12} = \frac{9}{12} = \frac{3}{4}$$

$U = 0'75 < 0'78 \rightarrow$ planificable

$T_M = mcm(60, 80, 120) = 240$

$20 \leq T_S \leq 60 \rightarrow T_S = 20$



56. $U = \frac{1}{5} + \frac{1}{10} + \frac{1}{10} + \frac{1}{2} + \frac{7}{100} = \frac{20+10+10+50+7}{100} = \frac{97}{100}$

RMS $\rightarrow U_0(5) = 5(\sqrt[5]{2} - 1) = 0'74$

$\rightarrow 0'97 > 0'74 \rightarrow$ no se puede afirmar ni descartar planificabilidad

EDF $\rightarrow 0'97 < 1 \rightarrow$ planificable

57. $U = \frac{1}{5} + \frac{1}{10} + \frac{1}{5} + \frac{1}{2} + \frac{7}{100} = \frac{20+10+20+50+7}{100} = \frac{107}{100}$